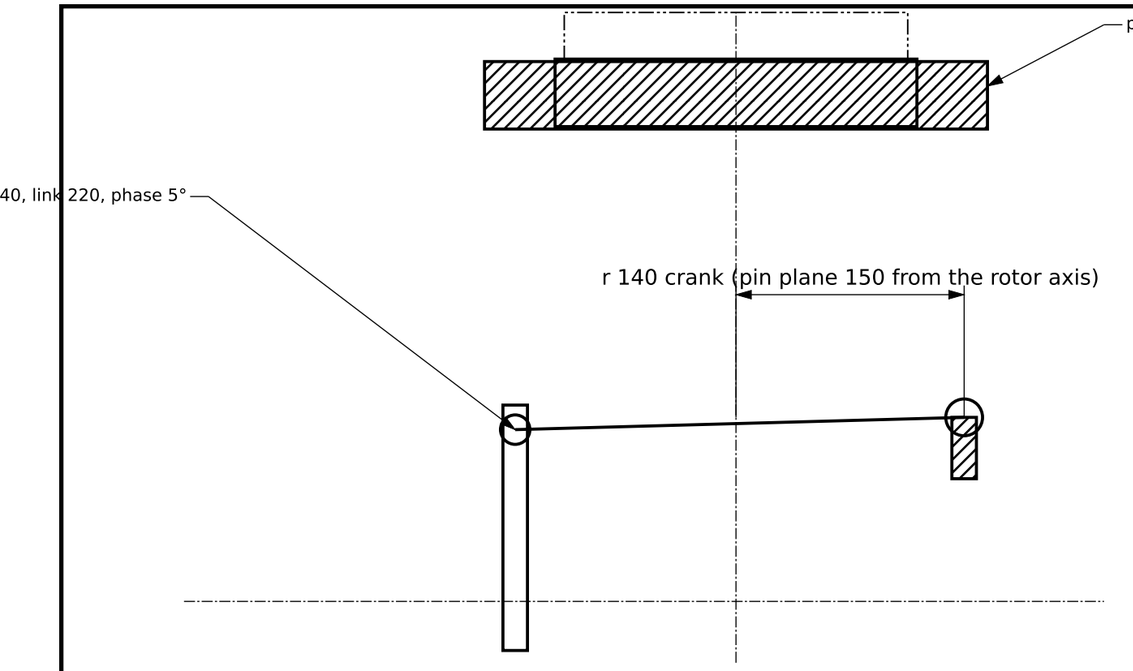
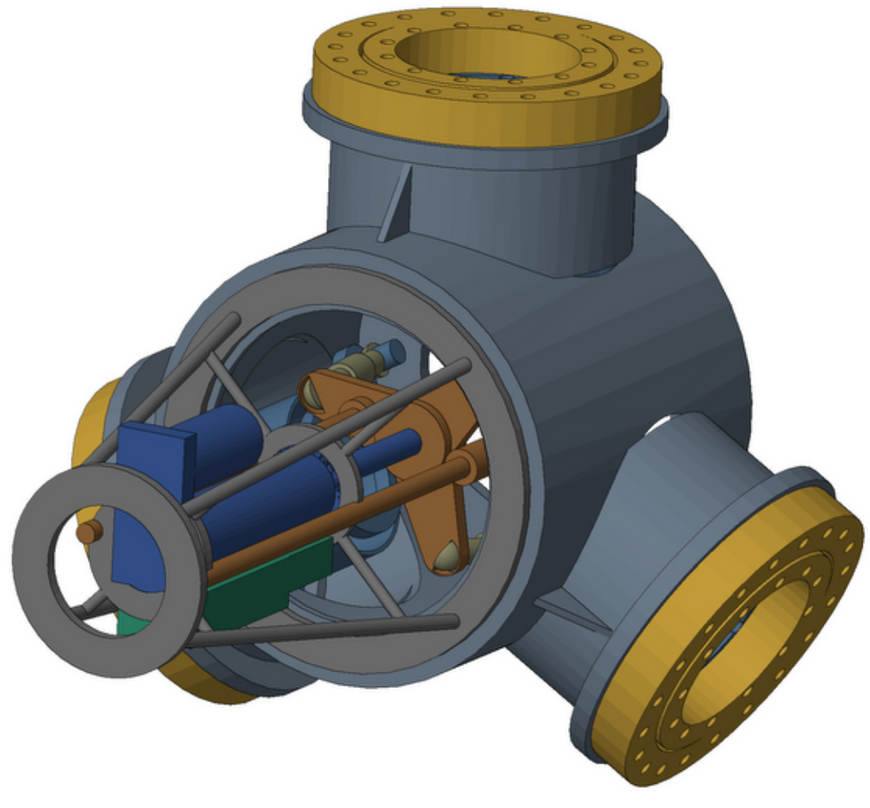
BLADE 1 PLANE, PITCH −2° (schematic)



AXONOMETRIC VIEW (not to scale)



Pitch system (hub_yaw_m8, crank r140)	Value
Cylinder stroke	204 mm
Force at trip (design)	23.5 kN
Rated force	28.5 kN
Motor peak torque	12.6 Nm
Energy per feather stroke	4.14 kJ
Strokes from one module (EOL)	3.6
Storm hold force on screw	37.1 kN
Motor brake needed	14.2 Nm
Pitch bearing	Dpw 300, balls 30, Z 28
Bearing s0 (SLM H)	1.37
Clash check	clean at −2, 10, 30, 45, 60, 80, 92°

NOTES

1. Kinematics from analysis/hub/collective.py with R_C 140, S_P 150, R_Y 140, L 220, phase 5° (cad/hub_m8/model_m8.py). The pin plane was moved out from 118 to 150 so the crank pin clears the drum bearing housing (r ≤ 120).
2. Bearings: 4-point ungeared, 20 × M16 outer to the neck flange (PCD 360), 12 × M16 inner to the blade root (PCD 245). Grease, sealed, −40 °C grease.
3. Cylinder ≈ 28.5 kN rated, 204 mm stroke, motor brake ≥ 14.2 Nm; supercap module sized for 5.0 feather strokes at end of life; nose frame carries both.
4. Mechanical end stops at −2.5° and 93° on crank 1; limit switches at −2° and 92°.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

COLLECTIVE PITCH · BEARINGS, CRANKS, LINKS, YOKE, CYLINDER

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-250	A · 2026-11-06	1:5
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see BOM	≈ 140 kg pitch parts (budget)	Claude / not checked